

TrES-5b

R-0706

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_251004 · created 2026-07-27T22:20:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460952.768 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.035
Transit depth (Rp/Rs)^2	3.3 +/- 1.28 [%]
Orbital Inclination (inc)	83.76 +/- 2.08
Airmass coefficient 1 (a1)	0.981 +/- 0.032
Airmass coefficient 2 (a2)	0.012 +/- 0.022
Scatter in the residuals of the lightcurve fit is	3.29 %
Best Comparison Star	#3 - [563, 338]
Optimal Method	PSF photometry
Transit Duration (day)	0.072 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.182 ± 0.035 vs 0.142 (deviation 1.14σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.26σ)
Mid-time O–C	6.0 min
Depth SNR	5.2
Residual scatter	3.29 %

Light curve

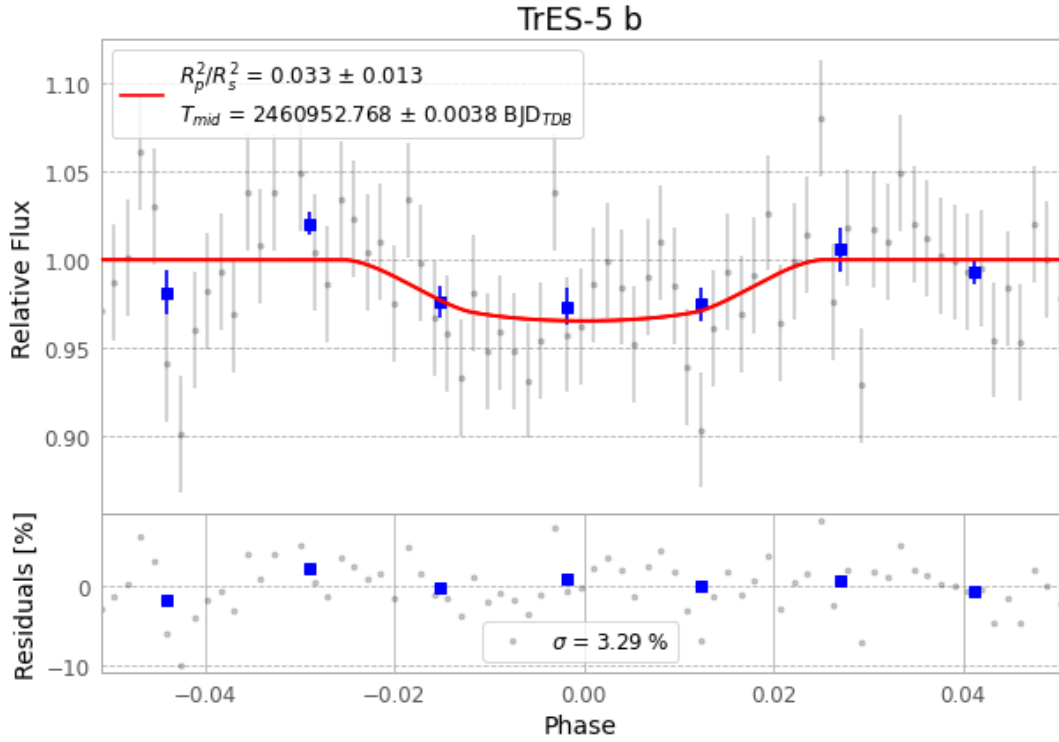


Figure 1 — FinalLightCurve_TrES-5 b_2025-10-03.png (SHA-256 9d59533c4265b41a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 324], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1d295fa4a04e859a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

73 FITS frames (48 MB), TRES-5251004043316.FITS → TRES-5251004080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-251004.log (1ffa82777ab9...), FinalLightCurve_TrES-5 b_2025-10-03.png (9d59533c4265...), FinalLightCurve_TrES-5 b_2025-10-03.csv (07ff4e8e6f90...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 22:20Z.